LayerSkip-Llama2-7B + llama-68M | E21 P1 K10 R4 W40 M15 C2 / P2 K5 R16 W20 M15 C3 | P1 hit -> P2 hit | UID 84_t0 | steps 7-8

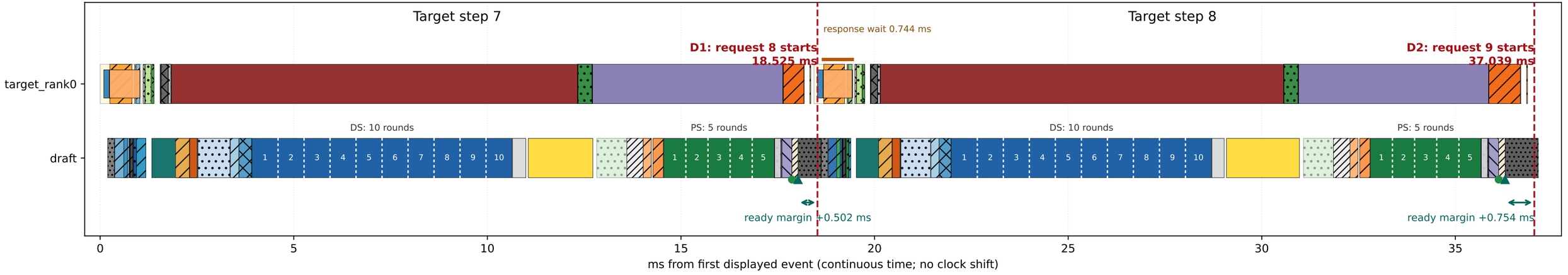
ver1.8 working tree; local RTX4090 PIX 4,5; FP16 B1 T0.7; 12 raw turns cap128; diagnostic | E21 P1 K10 R4 W40 M15 C2 / P2 K5 R16 W20 M15 C3 | B=1, K1/K2 10/5

Fused target graph_pre = layers + exit norm/head; graph_post = layers + final head. Head spans are included, not missing.

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



QUESTION MEANS over eligible diagnostic turns (not the two example steps); n = observed step count.

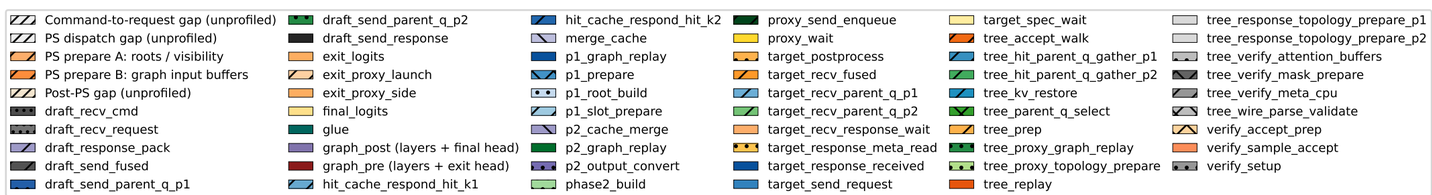
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	165	40.658	2.770	150.153
P2 hit	161	48.805	2.546	136.525
Miss	49	10.537	2.426	119.819
Overall	**375**	**89.463**	**2.605**	**138.721**

Rate, AL, latency and TPS: per-turn ratios -> within-question means -> question mean. Missing-source conditional metrics are undefined.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	165	18.466	0.201	3.770	+1.761	0.186	1.477	+0.547
P2 hit	161	18.664	0.202	3.816	+1.799	0.186	1.496	+0.565
Miss	49	20.338	0.195	4.958	+3.011	0.185	1.637	+0.712
Overall	**375**	**18.829**	**0.201**	**3.950**	**+1.944**	**0.186**	**1.501**	**+0.571**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip-Llama2-7B + llama-68M | E21 P1 K10 R4 W40 M15 C2 / P2 K5 R16 W20 M15 C3 | P2 hit -> P1 hit | UID 84_t0 | steps 55-56

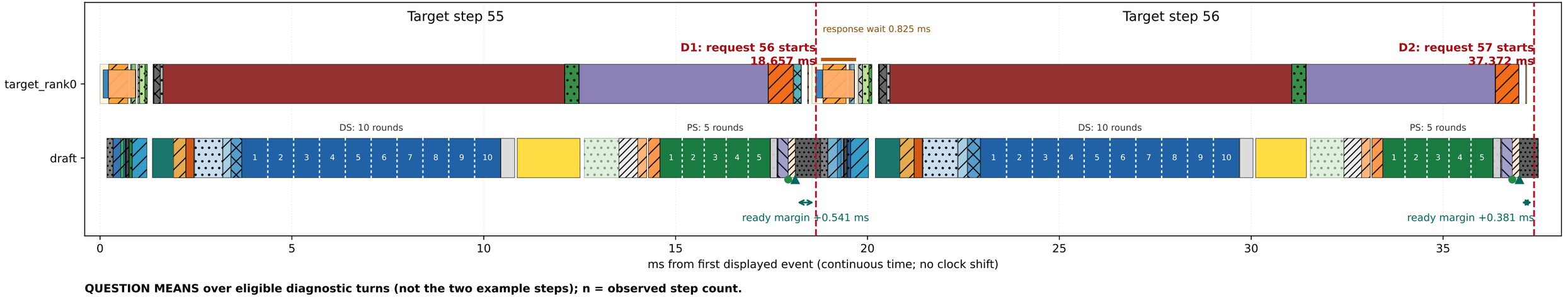
ver1.8 working tree; local RTX4090 PIX 4,5; FP16 B1 T0.7; 12 raw turns cap128; diagnostic | E21 P1 K10 R4 W40 M15 C2 / P2 K5 R16 W20 M15 C3 | B=1, K1/K2 10/5

Fused target graph_pre = layers + exit norm/head; graph_post = layers + final head. Head spans are included, not missing.

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

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QUESTION MEANS over eligible diagnostic turns (not the two example steps); n = observed step count.

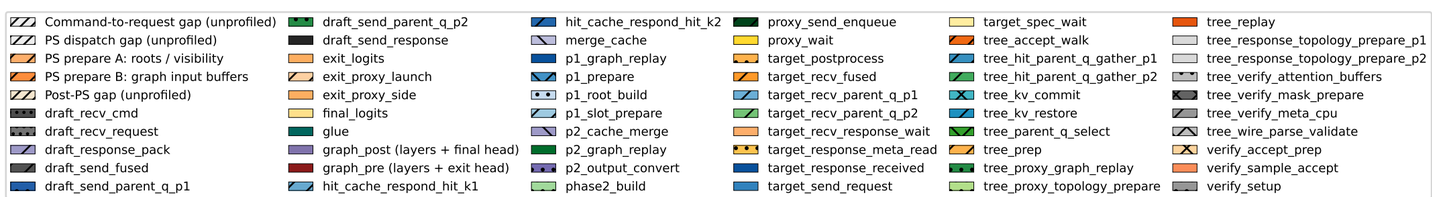
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
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Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip-Llama2-7B + llama-68M | E21 P1 K10 R4 W40 M15 C2 / P2 K5 R16 W20 M15 C3 | Miss -> P2 hit | UID 84_t0 | steps 36-37

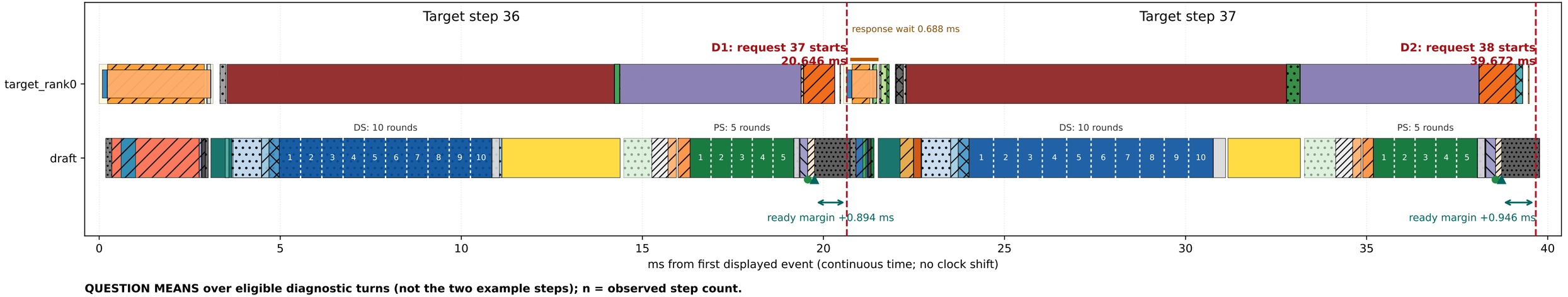
ver1.8 working tree; local RTX4090 PIX 4,5; FP16 B1 T0.7; 12 raw turns cap128; diagnostic | E21 P1 K10 R4 W40 M15 C2 / P2 K5 R16 W20 M15 C3 | B=1, K1/K2 10/5

Fused target graph_pre = layers + exit norm/head; graph_post = layers + final head. Head spans are included, not missing.

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Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post (layers + final head)
graph_pre (layers + exit head)
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_parent_q_gather_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_response_topology_prepare_p1
tree_response_topology_prepare_p2
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

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Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.